React Trend Application

Complete Setup and Operations Guide

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Project Repository: https://github.com/Surya-pkh/Projects/tree/main/react-trend-deployment

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Executive Summary

The React Trend Application is a production-ready e-commerce platform built with modern DevOps practices and deployed on AWS cloud infrastructure. This document provides comprehensive guidance for setup, deployment, monitoring, and ongoing operations.

Key Features

- **Modern React Frontend:** E-commerce interface with responsive design
- **Containerized Deployment:** Docker containers orchestrated with Kubernetes
- **Cloud Infrastructure:** AWS EKS cluster with auto-scaling capabilities
- **CI/CD Automation:** Jenkins pipeline with automated testing and deployment
- **Comprehensive Monitoring:** Prometheus and Grafana stack for observability
- **High Availability:** Multi-AZ deployment with load balancing

Production URLs

- **Application:** `http://a33f4d4d9655c4121a50242bb0a3942b-321113.us-west-2.elb.amazonaws.com`
- **Jenkins CI/CD:** `http://54.68.35.10:8080`
- **Grafana Monitoring:** `http://aa29945fe1c614be5ae0b521f91e2e87-1553642504.us-west-2.elb.amazonaws.com:3000`

Project Overview

Business Context

The React Trend Application (Trendify) is an e-commerce platform that provides:

- Product catalog browsing
- Shopping cart functionality
- User authentication workflows
- Responsive design for mobile and desktop

Technical Stack

- **Frontend:** React 18+ with Vite build system
- **Web Server:** nginx 1.29.1
- **Container Platform:** Docker + Kubernetes (AWS EKS)
- **Infrastructure:** AWS (VPC, EKS, EC2, ALB)
- **CI/CD:** Jenkins with automated pipelines
- **Monitoring:** Prometheus + Grafana
- **Container Registry:** Docker Hub

Deployment Model

- **Environment:** Production on AWS EKS
- **Scaling:** Horizontal Pod Autoscaler (2-10 pods)
- **Load Balancing:** AWS Application Load Balancer
- **High Availability:** Multi-AZ deployment across us-west-2a and us-west-2b

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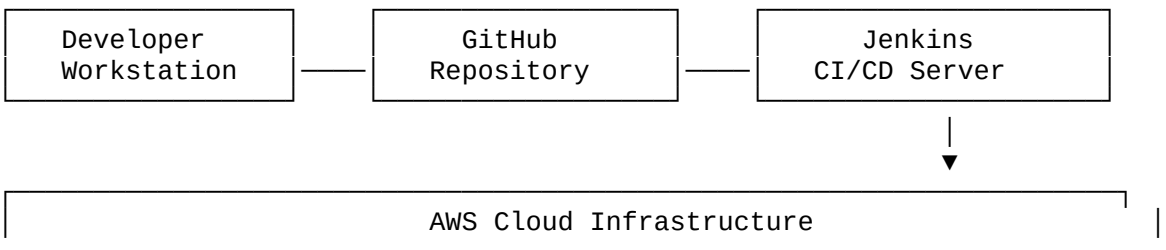
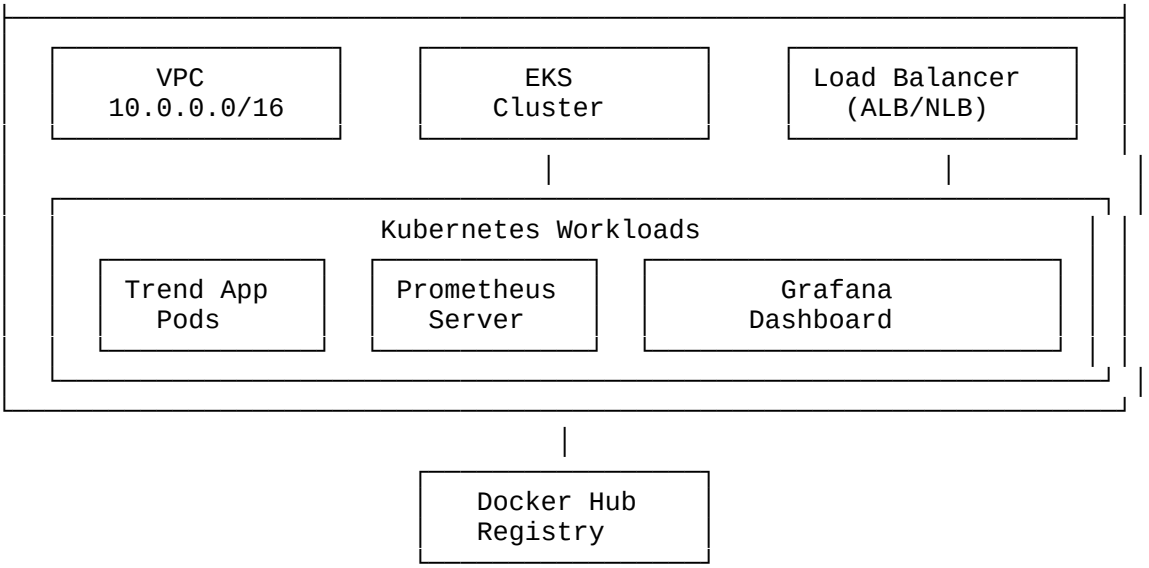
Architecture Overview

High-Level System Architecture





Network Architecture

VPC Design:

```
VPC: 10.0.0.0/16
├── Public Subnet 1: 10.0.1.0/24 (us-west-2a)
├── Public Subnet 2: 10.0.2.0/24 (us-west-2b)
├── Private Subnet 1: 10.0.3.0/24 (us-west-2a)
└── Private Subnet 2: 10.0.4.0/24 (us-west-2b)
```

Component Architecture

1. Frontend Application Layer

- **Technology:** React + Vite
- **Purpose:** E-commerce user interface
- **Features:** SPA with responsive design, product catalog, shopping cart

2. Web Server Layer

- **Technology:** nginx
- **Configuration:** Port 3000, health checks at `/health`, metrics at `/metrics`
- **Purpose:** Static file serving and routing with React Router support

3. Container Orchestration

- **Platform:** Kubernetes (AWS EKS)
- **Components:** Deployments, Services, ConfigMaps, HPA, Ingress

4. Infrastructure Layer

- **Provider:** AWS
- **Services:** EKS, VPC, ALB, EC2, IAM

Prerequisites

Required Tools and Versions

Tool	Version	Purpose
AWS CLI	v2.x	AWS resource management
kubectl	v1.27+	Kubernetes cluster management
Terraform	v1.5+	Infrastructure as Code
Docker	v20.x+	Container management
Git	Latest	Version control
Node.js	v18+	Local development (optional)

AWS Account Requirements

- **Permissions:** EC2, EKS, VPC, IAM full access
- **Credentials:** AWS Access Key and Secret Key configured
- **Key Pair:** EC2 key pair for SSH access to instances
- **Resource Limits:** Ensure sufficient VPC, EKS, and EC2 limits

External Account Requirements

- **Docker Hub:** Account with repository access
- **GitHub:** Repository access for source code

Local Environment Setup

```
# Verify AWS CLI
aws sts get-caller-identity

# Verify kubectl
kubectl version --client

# Verify terraform
terraform version

# Verify docker
docker version
```

Initial Setup

1. Repository Cloning

```
# Clone the main repository
git clone https://github.com/Surya-pkh/Projects.git
cd Projects/react-trend-deployment

# Verify project structure
ls -la
```

2. Configuration Setup

```
# Copy terraform variables template
cp terraform.tfvars.example terraform/terraform.tfvars
```

```
# Edit configuration file
nano terraform/terraform.tfvars
```

Required Configuration Values:

```
project_name = "trend-app"
environment  = "production"
region       = "us-west-2"
key_name     = "your-ec2-key-pair-name"
```

3. Docker Hub Configuration

Update the following files with your Docker Hub username:

- `Jenkinsfile` (line 45): `DOCKER_IMAGE = "your-username/trend-app"`
- `kubernetes/deployment.yaml` (line 20): Update image reference

4. Environment Verification

```
# Check AWS connectivity
aws eks list-clusters --region us-west-2

# Verify terraform syntax
cd terraform
terraform fmt
terraform validate
```

Infrastructure Deployment

Phase 1: Core Infrastructure Deployment

```
# Navigate to terraform directory
cd terraform

# Initialize terraform
terraform init

# Review planned changes
terraform plan -var-file="terraform.tfvars"

# Apply infrastructure changes
terraform apply -var-file="terraform.tfvars"
```

Expected Infrastructure Components:

- VPC with public/private subnets across 2 AZs
- EKS cluster with managed node groups
- Jenkins EC2 instance (t3.medium)
- Monitoring EC2 instance (t3.medium)
- Security groups and IAM roles
- Application Load Balancer

Estimated Deployment Time: 15-20 minutes

Phase 2: Kubernetes Configuration

```
# Update kubectl configuration
aws eks update-kubeconfig --region us-west-2 --name trend-app-cluster

# Verify cluster access
kubectl cluster-info
kubectl get nodes

# Check node status
kubectl get nodes -o wide
```

Phase 3: Namespace Creation

```
# Create application namespace
kubectl apply -f kubernetes/namespace.yaml

# Verify namespace creation
kubectl get namespaces
```

Infrastructure Verification Checklist:

- [ ] EKS cluster status is ACTIVE
- [ ] Worker nodes are in Ready state
- [ ] Jenkins instance is accessible via SSH
- [ ] Monitoring instance is running
- [ ] Load balancer is provisioned

Application Deployment

Method 1: Automated Deployment Script

```
# Make deployment script executable
chmod +x deploy.sh

# Deploy all components
./deploy.sh all

# Check deployment status
./deploy.sh status
```

Method 2: Manual Step-by-Step Deployment

Step 1: Build and Push Docker Image

```
# Build Docker image
docker build -t your-dockerhub-username/trend-app:latest .

# Test image locally
docker run -p 3000:3000 your-dockerhub-username/trend-app:latest

# Push to Docker Hub
docker push your-dockerhub-username/trend-app:latest
```

Step 2: Deploy Kubernetes Resources

```
# Deploy configuration
kubectl apply -f kubernetes/configmap.yaml

# Deploy application
kubectl apply -f kubernetes/deployment.yaml

# Expose application
kubectl apply -f kubernetes/service.yaml

# Configure auto-scaling
kubectl apply -f kubernetes/hpa.yaml

# Optional: Configure ingress
kubectl apply -f kubernetes/ingress.yaml
```

Step 3: Verify Deployment

```
# Check pod status
kubectl get pods -n trend

# Check service status
kubectl get services -n trend

# Check deployment status
kubectl get deployments -n trend

# Check horizontal pod autoscaler
kubectl get hpa -n trend
```

Step 4: Access Application

```
# Get load balancer URL
kubectl get service trend-app-service -n trend

# Test application health
curl -I http://<LOAD_BALANCER_URL>/health

# Access application
open http://<LOAD_BALANCER_URL>
```

Method 3: CI/CD Pipeline Deployment (Recommended for Production)

Jenkins Setup and Configuration

1. **Access Jenkins:**

   ```
   # Get Jenkins IP from terraform output
   terraform output jenkins_public_ip

   # Access via browser: http://<jenkins-ip>:8080
   ```

2. **Initial Setup:**

   ```
   # Get initial admin password
   ssh -i your-key.pem ubuntu@<jenkins-ip>
   sudo cat /var/lib/jenkins/secrets/initialAdminPassword
   ```

3. **Configure Credentials:**

- Navigate to Manage Jenkins → Manage Credentials
- Add Docker Hub credentials (ID: `dockerhub-credentials`)
- Add AWS credentials (ID: `aws-credentials`)

4. **Create Pipeline:**
    - New Item → Pipeline
    - Pipeline script from SCM
    - Repository URL: Your GitHub repository
    - Script Path: `Jenkinsfile`

Pipeline Execution

The Jenkins pipeline includes the following stages:

1. **Checkout** - Clone repository from GitHub
2. **Lint & Security** - Code quality and security checks
3. **Build React App** - npm build process
4. **Build Docker Image** - Container creation
5. **Test Docker Image** - Container functionality tests
6. **Push to Registry** - Upload to Docker Hub
7. **Deploy to EKS** - Kubernetes deployment
8. **Verify Deployment** - Health checks and smoke tests

Pipeline Trigger Options:

- **Manual:** Trigger from Jenkins UI
- **Webhook:** Automatic trigger on Git push
- **Scheduled:** Cron-based execution

Monitoring Setup

Phase 1: Monitoring Infrastructure

The monitoring stack is automatically deployed with the infrastructure and includes:

- **Prometheus Server:** Metrics collection and storage
- **Grafana Dashboard:** Data visualization
- **Node Exporter:** System metrics collection

Phase 2: Access Configuration

```
# Get monitoring instance IP
terraform output monitoring_public_ip

# Access URLs:
# Prometheus: http://<monitoring-ip>:9090
# Grafana: http://<monitoring-ip>:3000
```

Phase 3: Grafana Dashboard Setup

1. **Initial Login:**

    - URL: `http://aa29945fe1c614be5ae0b521f91e2e87-1553642504.us-west-2.elb.amazonaws.com:3000`
    - Username: `admin`
    - Password: `admin123`
2. **Add Prometheus Data Source:**
    - Navigate to Configuration → Data Sources
    - Add Prometheus
    - URL: `http://prometheus.monitoring.svc.cluster.local:9090`
    - Save & Test
3. **Import Dashboards:**
    - Use dashboard IDs: 15757 (Kubernetes), 8588 (Node Exporter), 3662 (Prometheus)
    - Or import custom dashboards from `monitoring/` directory

Key Metrics Monitored

Application Metrics

- **Availability:** `up{job="kubernetes-pods", pod=~"trend-app.*"}`
- **Request Rate:** `rate(nginx_http_requests_total[5m])`
- **Error Rate:** `rate(nginx_http_requests_total{status=~"5.."}[5m])`
- **Response Time:** `histogram_quantile(0.95, nginx_http_request_duration_seconds_bucket)`

Infrastructure Metrics

- **CPU Usage:** `rate(container_cpu_usage_seconds_total[5m]) * 100`
- **Memory Usage:** `container_memory_usage_bytes / 1024 / 1024`
- **Network I/O:** `rate(container_network_receive_bytes_total[5m])`
- **Pod Restarts:** `increase(kube_pod_container_status_restarts_total[1h])`

Kubernetes Metrics

- **Pod Status:** `kube_pod_status_phase`
- **Node Status:** `kube_node_status_condition`
- **Deployment Health:** `kube_deployment_status_replicas_available`

CI/CD Pipeline

Pipeline Architecture

The Jenkins pipeline implements a complete CI/CD workflow with the following characteristics:

Pipeline Features

- **Declarative Syntax:** Easy to read and maintain

- **Multi-Stage:** Comprehensive build and deployment process
- **Security Scanning:** Vulnerability assessment with Trivy
- **Parallel Execution:** Optimized build times
- **Error Handling:** Proper cleanup and notification

Jenkinsfile Structure

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = 'your-username/trend-app'
        KUBE_NAMESPACE = 'trend'
        HEALTH_CHECK_URL = 'http://localhost:3000/health'
    }

    stages {
        stage('Checkout') { ... }
        stage('Lint & Security Scan') { ... }
        stage('Build React App') { ... }
        stage('Build Docker Image') { ... }
        stage('Test Docker Image') { ... }
        stage('Push to Docker Hub') { ... }
        stage('Deploy to EKS') { ... }
        stage('Verify Deployment') { ... }
        stage('Smoke Tests') { ... }
    }
}
```

Pipeline Stages in Detail

1. Checkout Stage

```
stage('Checkout') {
    steps {
        git branch: 'main', url: 'https://github.com/your-username/trend-app'
        sh 'ls -la'
    }
}
```

2. Security and Quality Stage

```
stage('Lint & Security Scan') {
    parallel {
        stage('Lint Code') {
            steps {
                sh 'npm install'
                sh 'npm run lint || true'
            }
        }
        stage('Security Scan') {
            steps {
                sh 'trivy fs . || true'
            }
        }
    }
}
```

3. Build and Test Stages

```
stage('Build React App') {
    steps {
        sh 'npm install'
        sh 'npm run build'
    }
}

stage('Build Docker Image') {
    steps {
        script {
            docker.build("${DOCKER_IMAGE}:${BUILD_NUMBER}")
        }
    }
}
```

4. Deployment Stage

```
stage('Deploy to EKS') {
    steps {
        withAWS(credentials: 'aws-credentials') {
            sh 'aws eks update-kubeconfig --name trend-app-cluster --region us-
west-2'
            sh 'kubectl set image deployment/trend-app trend-app=$
{DOCKER_IMAGE}:${BUILD_NUMBER} -n ${KUBE_NAMESPACE}'
            sh 'kubectl rollout status deployment/trend-app -n $
{KUBE_NAMESPACE}'
        }
    }
}
```

Pipeline Triggering

1. Manual Execution

- Access Jenkins UI
- Navigate to trend-app pipeline
- Click "Build Now"

2. Webhook Integration

```
# Configure GitHub webhook
# URL: http://<jenkins-ip>:8080/github-webhook/
# Content type: application/json
# Events: Push events
```

3. Scheduled Execution

```
triggers {
    cron('H 2 * * *') // Daily at 2 AM
}
```

Operational Procedures

Daily Operations

Health Check Procedures

```
# Check application health
curl -I http://a33f4d4d9655c4121a50242bb0a3942b-321113.us-
west-2.elb.amazonaws.com/health

# Check pod status
kubectl get pods -n trend

# Check resource usage
kubectl top pods -n trend
kubectl top nodes

# Check recent logs
kubectl logs -n trend deployment/trend-app --tail=50
```

Morning Health Check Checklist

- [ ] Application is accessible via load balancer
- [ ] All pods are in Running state
- [ ] No recent error spikes in monitoring
- [ ] Jenkins pipeline last run was successful
- [ ] Resource utilization is within normal ranges

Weekly Operations

Resource Review

```
# Check resource utilization trends
kubectl top pods -n trend
kubectl top nodes

# Review HPA metrics
kubectl get hpa -n trend -o wide

# Check for any pending pods
kubectl get pods -n trend | grep Pending
```

Security Updates

```
# Update Docker base images
docker pull node:18-alpine
docker pull nginx:alpine

# Rebuild and deploy
docker build -t your-username/trend-app:$(date +%Y%m%d) .
```

Monthly Operations

Infrastructure Review

```
# Review AWS costs
aws ce get-cost-and-usage --time-period Start=2025-08-01,End=2025-08-31 --
granularity MONTHLY --metrics BlendedCost
```

```
# Review EKS cluster version
aws eks describe-cluster --name trend-app-cluster --region us-west-2

# Review security groups
aws ec2 describe-security-groups --filters "Name=group-name,Values=*trend*"
```

Scaling Operations

Manual Scaling

```
# Scale application pods
kubectl scale deployment trend-app --replicas=5 -n trend

# Scale cluster nodes (if needed)
aws eks update-nodegroup-config --cluster-name trend-app-cluster --nodegroup-
name workers --scaling-config minSize=2,maxSize=6,desiredSize=3
```

Auto-Scaling Configuration

```
# HPA configuration
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: trend-app-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: trend-app
  minReplicas: 2
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 70
```

Monitoring and Alerting

Prometheus Configuration

Key Scraping Targets

```
scrape_configs:
  - job_name: 'kubernetes-pods'
    kubernetes_sd_configs:
    - role: pod
    relabel_configs:
    - source_labels: [__meta_kubernetes_pod_annotation_prometheus_io_scrape]
      action: keep
      regex: true
```

Important Metrics

Application Health Metrics:

- `nginx_up`: Service availability status
- `nginx_requests_total`: Total request count
- `nginx_request_duration_seconds`: Request latency distribution

Resource Utilization Metrics:

- `container_cpu_usage_seconds_total`: CPU usage over time
- `container_memory_usage_bytes`: Current memory usage
- `kube_pod_container_resource_limits`: Resource limits

Kubernetes Cluster Metrics:

- `kube_pod_status_phase`: Pod lifecycle status
- `kube_deployment_status_replicas`: Deployment replica status
- `kube_node_status_condition`: Node health status

Grafana Dashboard Configuration

Dashboard 1: Application Overview

```
{
  "title": "Trend App - Application Overview",
  "panels": [
    {
      "title": "Application Health Status",
      "type": "stat",
      "targets": [{
        "expr": "up{job=\"kubernetes-pods\", pod=~\"trend-app.*\"}"
      }]
    },
    {
      "title": "Request Rate",
      "type": "graph",
      "targets": [{
        "expr": "rate(nginx_http_requests_total[5m])"
      }]
    }
  ]
}
```

Dashboard 2: Infrastructure Metrics

```
{
  "title": "Trend App - Infrastructure",
  "panels": [
    {
      "title": "Pod CPU Usage",
      "type": "graph",
      "targets": [{
        "expr": "rate(container_cpu_usage_seconds_total{pod=~\"trend-app.*\"}
[5m]) * 100"
      }]
    },
    {
      "title": "Pod Memory Usage",
```

```
      "type": "graph",
      "targets": [{
        "expr": "container_memory_usage_bytes{pod=~\"trend-app.*\"} / 1024 /
1024"
      }]
    }
  ]
}
```

Alerting Rules

Critical Alerts

```
groups:
- name: trend-app-critical
  rules:
  - alert: ApplicationDown
    expr: up{job="kubernetes-pods", pod=~"trend-app.*"} == 0
    for: 1m
    labels:
      severity: critical
    annotations:
      summary: "Trend application is down"

  - alert: HighErrorRate
    expr: rate(nginx_http_requests_total{status=~"5.."}[5m]) /
rate(nginx_http_requests_total[5m]) > 0.05
    for: 5m
    labels:
      severity: critical
    annotations:
      summary: "High error rate detected"
```

Warning Alerts

```
- alert: HighCPUUsage
  expr: rate(container_cpu_usage_seconds_total{pod=~"trend-app.*"}[5m]) * 100 >
80
  for: 2m
  labels:
    severity: warning
  annotations:
    summary: "High CPU usage detected"

- alert: HighMemoryUsage
  expr: container_memory_usage_bytes{pod=~"trend-app.*"} /
container_spec_memory_limit_bytes > 0.8
  for: 2m
  labels:
    severity: warning
  annotations:
    summary: "High memory usage detected"
```

Troubleshooting Guide

Common Issues and Solutions

1. Application Not Accessible

Symptoms:

- HTTP timeout when accessing application URL
- Load balancer health checks failing

Diagnosis Steps:

```
# Check pod status
kubectl get pods -n trend

# Check service endpoints
kubectl get endpoints -n trend

# Check load balancer status
kubectl describe service trend-app-service -n trend

# Test internal connectivity
kubectl exec -n trend deployment/trend-app -- curl localhost:3000/health
```

Solutions:

```
# Restart pods if unhealthy
kubectl rollout restart deployment/trend-app -n trend

# Check and fix service selector
kubectl edit service trend-app-service -n trend

# Verify security group rules allow traffic
aws ec2 describe-security-groups --group-ids sg-xxxxx
```

2. Pods Stuck in Pending State

Symptoms:

- Pods show Pending status
- Unable to schedule on nodes

Diagnosis Steps:

```
# Describe pending pods
kubectl describe pod <pod-name> -n trend

# Check node resources
kubectl describe nodes

# Check node status
kubectl get nodes -o wide
```

Solutions:

```
# Scale cluster if resource constrained
aws eks update-nodegroup-config --cluster-name trend-app-cluster --nodegroup-
name workers --scaling-config desiredSize=3

# Check and remove taints if necessary
```

```
kubectl taint nodes <node-name> key=value:NoSchedule-
```

3. Jenkins Pipeline Failures

Symptoms:

- Build fails during deployment stage
- Authentication errors with AWS or Docker Hub

Common Solutions:

```
# Verify Jenkins credentials
# Go to Manage Jenkins → Manage Credentials

# Check AWS CLI configuration on Jenkins
ssh -i your-key.pem ubuntu@<jenkins-ip>
aws sts get-caller-identity

# Verify Docker Hub connectivity
docker login
docker push test-image
```

4. Monitoring Data Missing

Symptoms:

- Grafana shows no data
- Prometheus targets down

Diagnosis Steps:

```
# Check Prometheus targets
kubectl port-forward service/prometheus 9090:9090 -n monitoring
# Visit http://localhost:9090/targets

# Check Prometheus logs
kubectl logs deployment/prometheus -n monitoring

# Verify Grafana data source
# Check Grafana → Configuration → Data Sources
```

Solutions:

```
# Restart monitoring components
kubectl rollout restart deployment/prometheus -n monitoring
kubectl rollout restart deployment/grafana -n monitoring

# Verify network connectivity
kubectl exec -n monitoring deployment/prometheus -- nslookup
kubernetes.default.svc.cluster.local
```

Emergency Procedures

Application Emergency Response

Critical Outage Response:

1. **Immediate Assessment:**

   ```
   # Check overall system status
   ```

```
kubectl get pods -n trend
kubectl get nodes
kubectl get services -n trend
```

2. **Quick Recovery:**

```
# Restart application if pods are failing
kubectl rollout restart deployment/trend-app -n trend

# Scale up if resource issue
kubectl scale deployment trend-app --replicas=5 -n trend
```

3. **Rollback if Recent Deploy:**

```
# Check deployment history
kubectl rollout history deployment/trend-app -n trend

# Rollback to previous version
kubectl rollout undo deployment/trend-app -n trend
```

Infrastructure Emergency Response

EKS Cluster Issues:

```
# Check cluster status
aws eks describe-cluster --name trend-app-cluster --region us-west-2

# Check node group status
aws eks describe-nodegroup --cluster-name trend-app-cluster --nodegroup-name
workers --region us-west-2

# Force node replacement if needed
aws eks update-nodegroup-config --cluster-name trend-app-cluster --nodegroup-
name workers --force-update-version --region us-west-2
```

Performance Issues

High Response Time

Diagnosis:

```
# Check resource utilization
kubectl top pods -n trend
kubectl top nodes

# Check HPA status
kubectl get hpa -n trend

# Check network latency
kubectl exec -n trend deployment/trend-app -- ping google.com
```

Solutions:

```
# Increase resources if constrained
kubectl patch deployment trend-app -n trend -p '{"spec":{"template":{"spec":
{"containers":[{"name":"trend-app","resources":{"limits":
{"cpu":"500m","memory":"1Gi"}}}]}}}}'

# Scale horizontally
kubectl scale deployment trend-app --replicas=5 -n trend
```

```
# Check and optimize nginx configuration
kubectl edit configmap nginx-config -n trend
```

Maintenance and Operations

Regular Maintenance Schedule

Daily Tasks (Automated via Monitoring)

- [ ] Application health verification
- [ ] Resource utilization monitoring
- [ ] Error rate monitoring
- [ ] Performance metrics review

Weekly Tasks

```
# 1. Security updates
# Check for base image updates
docker pull node:18-alpine
docker pull nginx:alpine

# 2. Resource optimization
kubectl top pods -n trend --sort-by=cpu
kubectl top pods -n trend --sort-by=memory

# 3. Log rotation and cleanup
kubectl logs -n trend deployment/trend-app --tail=1000 > trend-app-logs-$(date +
%Y%m%d).log

# 4. Backup verification
terraform show -json > terraform-state-backup-$(date +%Y%m%d).json
```

Monthly Tasks

```
# 1. Infrastructure cost review
aws ce get-cost-and-usage --time-period Start=2025-08-01,End=2025-08-31 --
granularity MONTHLY --metrics BlendedCost

# 2. Security audit
aws eks describe-cluster --name trend-app-cluster --region us-west-2 --query
'cluster.logging'

# 3. Performance optimization
# Review HPA metrics and adjust thresholds
kubectl edit hpa trend-app-hpa -n trend

# 4. Update cluster components
aws eks update-cluster-version --name trend-app-cluster --version 1.28 --region
us-west-2
```

Quarterly Tasks

- Security vulnerability assessment
- Disaster recovery testing
- Documentation updates
- Architecture review and optimization

- Cost optimization analysis

Backup and Recovery

Data Backup Strategy

```
# 1. Terraform state backup
cp terraform.tfstate terraform.tfstate.backup.$(date +%Y%m%d)

# 2. Kubernetes resource backup
kubectl get all -n trend -o yaml > trend-k8s-backup-$(date +%Y%m%d).yaml
```